Analisis de Salarios

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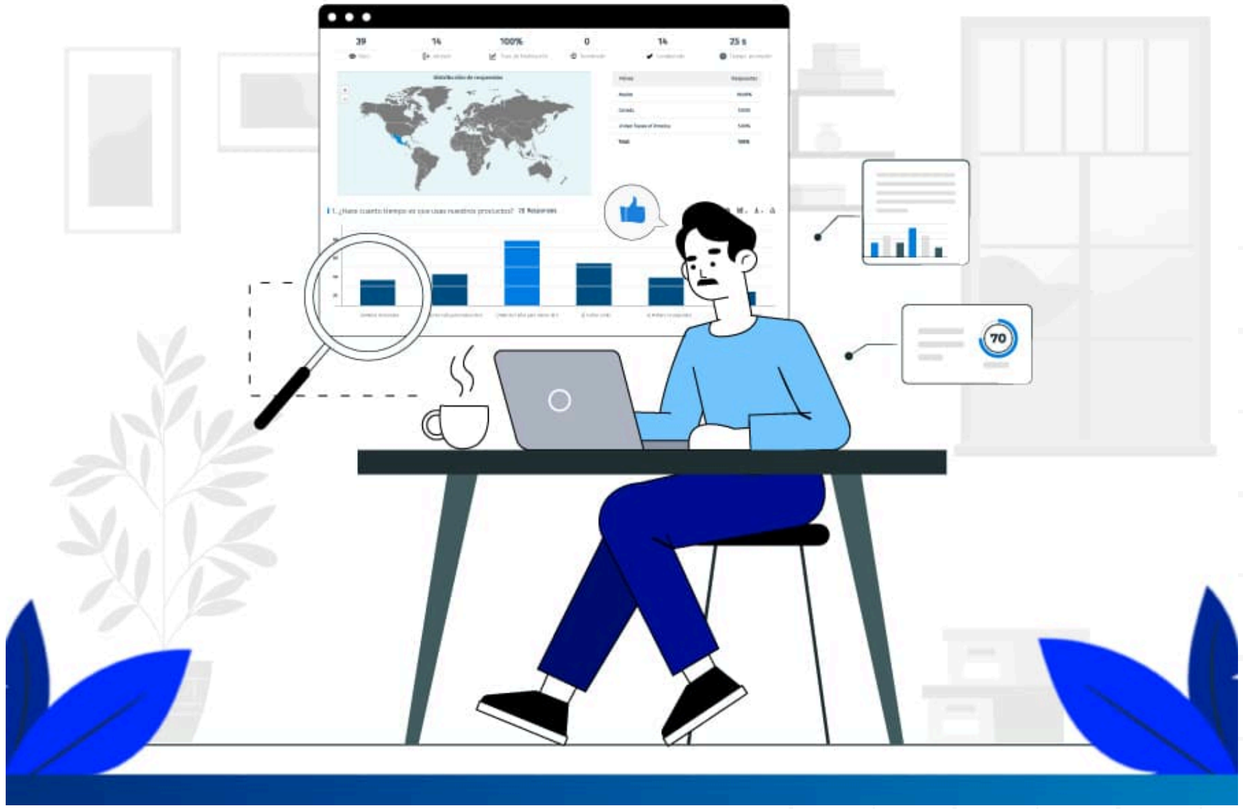
Data set

- Fuente: Kaggle (Data Science Job Salaries), de ai-jobs.net
- Tamaño: 607 registros, años 2020–2022
- Target: salary_in_usd
- Variables clave: experience_level, job_title, company_location, employee_residence, remote_ratio, company_size



Preparación de datos y transformaciones

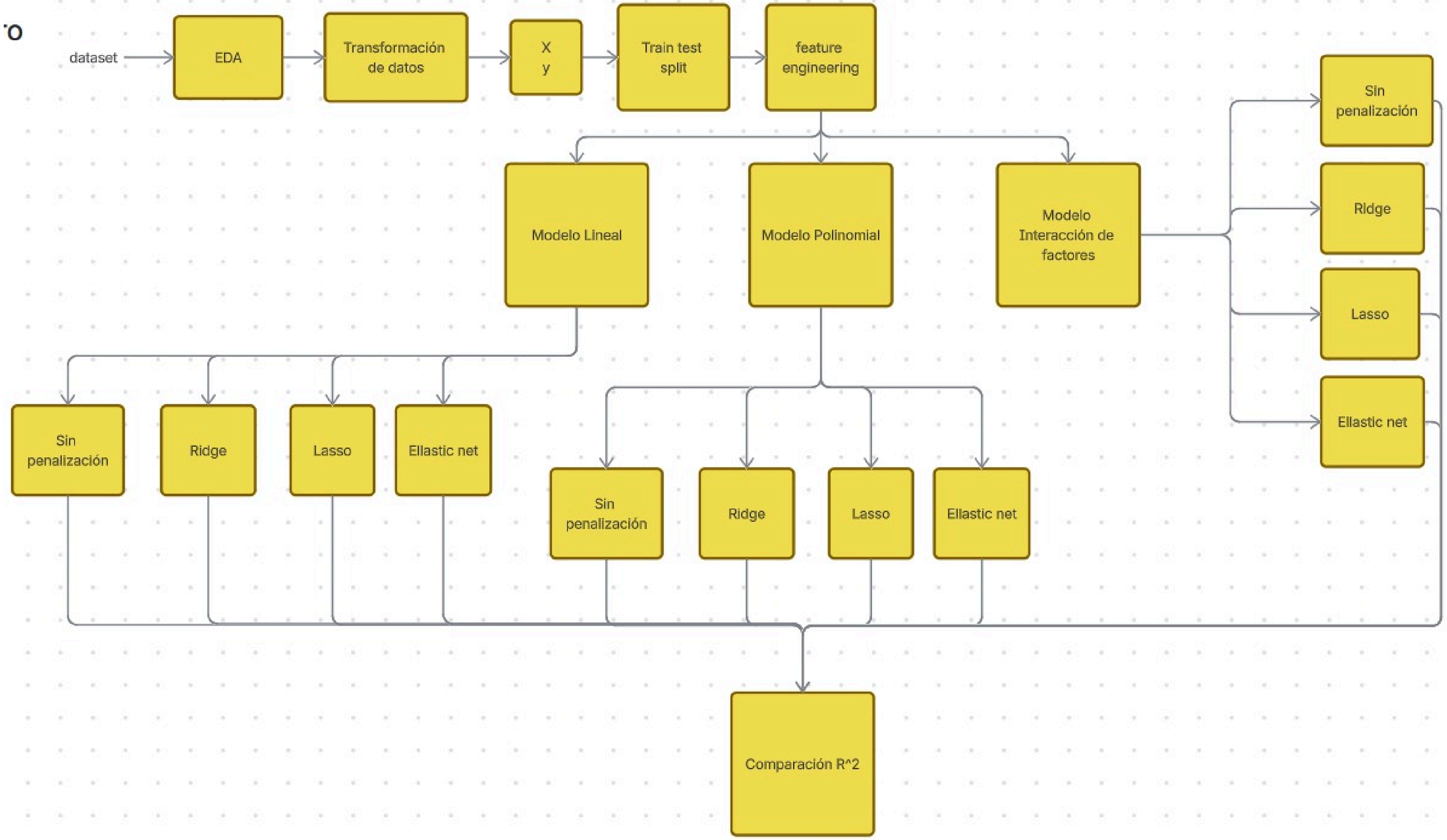
Limpieza:

- Se eliminan salary y salary_currency

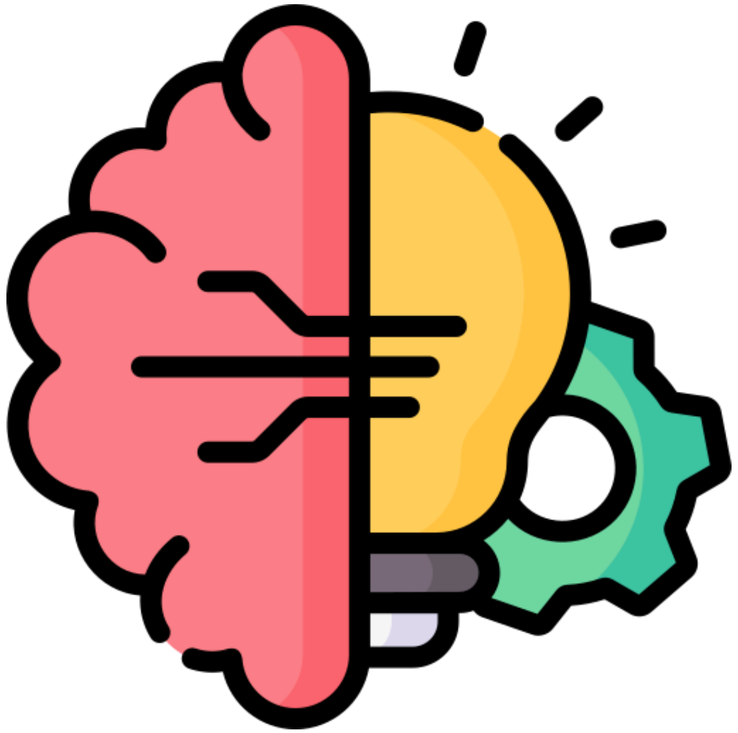
Dummies / One-hot encoding para categóricas:

- experience_level, employment_type, job_title,
- employee_residence, company_location, company_size.

Sacamos las 15 variables con mas correlación y las usamos en el modelo (esto para evitar overfitting)


dataset
EDA
Transformación de datos
X
y
Train test split
feature engineering
Modelo Lineal
Modelo Polinomial
Modelo Interacción de factores
Sin penalización
Ridge
Lasso
Ellastic net
Sin penalización
Ridge
Lasso
Ellastic net
Sin penalización
Ridge
Lasso
Ellastic net
Comparación R^2

Modelos propuestos



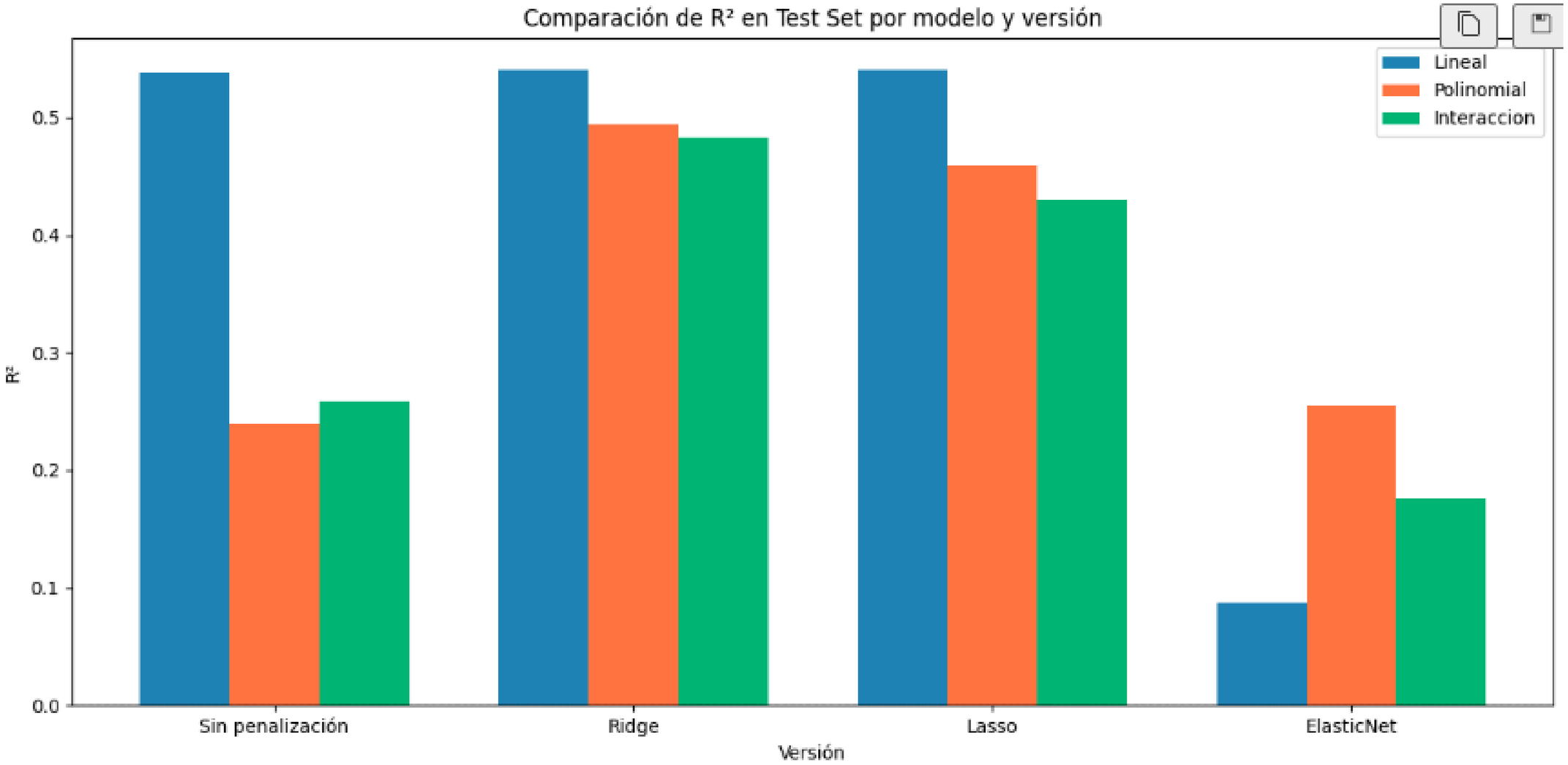
Modelos:

- Modelo 1: Lineal
- Modelo 2: Polinomial grado 2
- Modelo 3: Interacciones (grado 2, solo interacciones)

Cada uno con 4 versiones:

- Sin penalización
- Ridge (α=50)
- Lasso (α=50)
- ElasticNet (α=50)

Resultados (R^2) + comparación



P Value Factores significativos (p < 0.05)

	coef	std err	t	P>|t|	[0.025	0.975]
Intercepto	1.142e+05	2210.625	51.639	0.000	1.1e+05	1.18e+05
employee_residence_US	2.764e+04	6282.453	4.399	0.000	1.53e+04	4e+04
company_location_US	4634.7896	6150.786	0.754	0.452	-7452.752	1.67e+04
experience_level_SE	2.08e+04	3843.888	5.411	0.000	1.32e+04	2.84e+04
experience_level_EX	2.36e+04	2866.136	8.235	0.000	1.8e+04	2.92e+04
job_title_Principal Data Engineer	1.171e+04	2251.400	5.202	0.000	7288.315	1.61e+04
work_year	-9621.9048	5162.982	-1.864	0.063	-1.98e+04	524.404
job_title_Data Analytics Lead	1.111e+04	2219.171	5.006	0.000	6748.874	1.55e+04
Unnamed: 0	1.002e+04	4988.990	2.007	0.045	210.766	1.98e+04
job_title_Principal Data Scientist	9499.0759	2265.874	4.192	0.000	5046.174	1.4e+04
remote_ratio	238.4492	2322.613	0.103	0.918	-4325.957	4802.855

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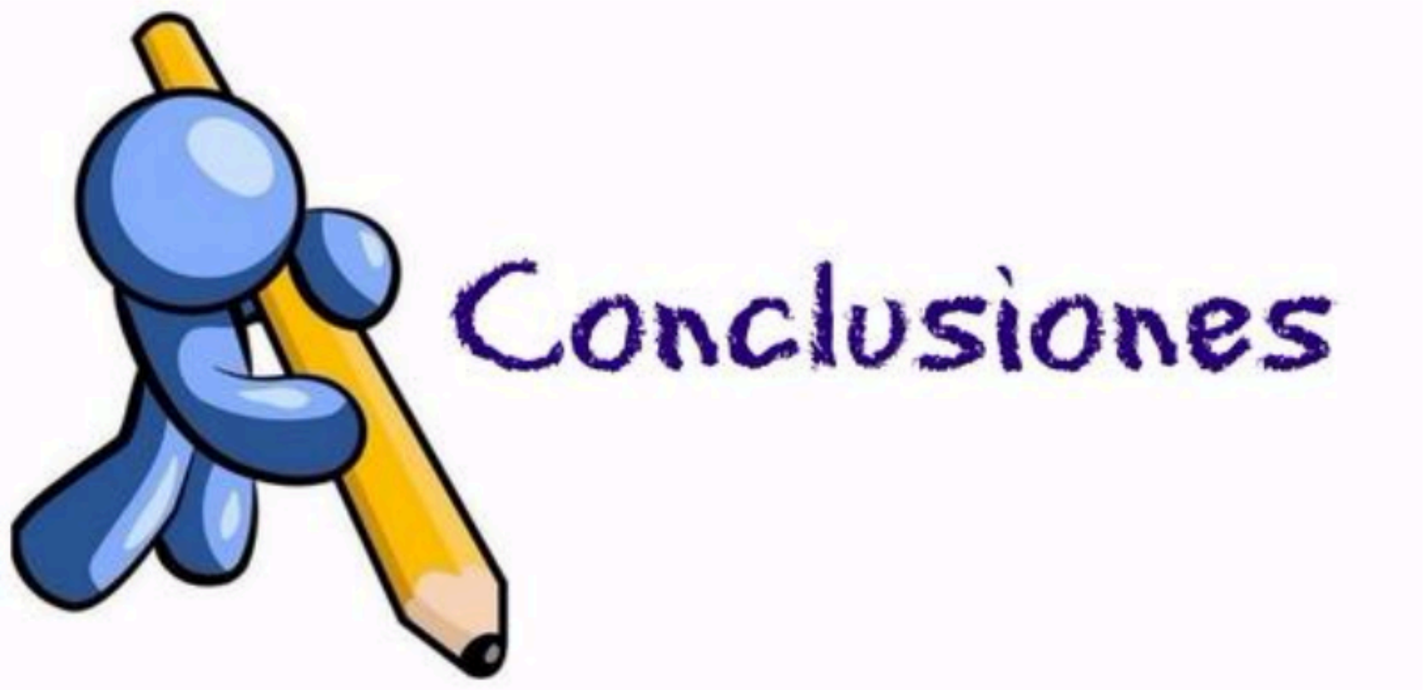

El mejor desempeño lo obtuvo la regresión lineal con Lasso (R^2 = 0.54 en test).

El salario se explica principalmente por:

- Residir en Estados Unidos
- Nivel de experiencia Senior (SE) y Ejecutivo (EX)
- Job titles específicos

Gracias